MIL-T-50611(MU)
6 May 1970

MILITARY SPECIFICATION

TRINITRORESORCINOL (STYPHNIC ACID)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers trinitroresorcinol for use in the manufacture of lead styphnate.

1.2 Classification. The trinitroresorcinol shall be of the following types:

Type I - Cream color (see table I)
Type II - Reddish-brown (see table I)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

RR-S-366 - Sieves, Test
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of
PPP-D-723 - Drums, Fiber

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

FSC 1376

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D1193-66 - Reagent Water

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606).

3. REQUIREMENTS

3.1 Material. The materials used in the manufacture of the trinitroresorcinol shall be of high quality. The trinitroresorcinol shall conform to the applicable requirements as specified herein and shall be visibly free of contaminants and foreign material such as grit and dirt.

3.2 Chemical and physical properties. The chemical and physical properties of trinitroresorcinol shall conform to the requirements of table I. The material shall be identified as being trinitroresorcinol (i.e., 2,4,6-trinitroresorcinol) by infrared spectroscopy.

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Table I. Chemical and physical properties

Property	Requirements Type I	Type II	Test par.
Appearance	Cream-colored powder	Reddish-brown powder	4.5.1
Moisture, percent	21-27	21-27	
Assay by determination of NO_2 groups, min., percent	98.0	98.0	4.5.2
Assay by titration with standard alkali, min, percent	99.0	99.0	4.5.3
Melting point (final), °C	Greater than 170°C Less than 180°C	Greater than 170°C Less than 178°C	4.5.4
Residue on ignition, max. percent	0.07	0.07	4.5.5
Water-insoluble matter, max. percent	0.02	0.02	4.5.6
Alcohol-insoluble, max. percent	0.01	0.01	4.5.7
Sulfate, including resorcinol disulfonic acid (as H_2SO_4), max. percent	1.00	1.30	4.5.8
Granulation, through No. 40 sieve, min. percent	100	100	4.5.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Acceptance inspection. Conformance of the trinitroresorcinol to the requirements of this specification shall be determined entirely by means of acceptance inspection. The acceptance inspection shall consist of an examination for acceptability of facilities and quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.2) for all of the requirements of this specification, and an examination of the sample of filled containers (4.3.3) for conformance to section 5.

4.3 Sampling.

4.3.1 Lot. A lot shall consist of all trinitroresorcinol of the same type, manufactured by the same process, packed as specified in the contract or order (see 6.2) and offered for acceptance at one time.

4.3.2 Sample for test. From each lot offered for acceptance under contract, a random sample shall be selected by the inspector in accordance with MIL-STD-105, inspection level II. A primary sample of one pint of material shall be removed from each container selected and stored in a wide mouth glass stoppered bottle. Each primary sample shall be labeled so that the container from which it was taken can be easily identified. A composite sample shall be made of equal portions of these samples and a minimum of one pint shall be used for acceptance tests as specified in 4.4 and 4.5. The sample shall be labeled to show the name and address of the manufacturer, plant, name of product, specification number, lot number, date of sampling, and contract number.

4.3.3 Sample for examination of filled containers. A random sample of filled containers shall be selected from each lot of trinitroresorcinol offered for acceptance under contract in accordance with MIL-STD-105, inspection level I, and acceptable quality level (AQL) = 2.5 percent defective, to determine compliance with the requirements of section 5.

4.4 Inspection methods. Unless otherwise specified, all examinations and tests shall be conducted as specified herein to determine compliance with the applicable requirements of section 3. Failure of any test by any specimen shall be cause for rejection of the lot represented. Water in accordance with ASTM D1193 and reagent chemicals shall be used. Any deviations from the test and/or sampling procedures specified herein shall require prior approval of the procuring activity.

4.4.1 Visual inspection. Conformance of the trinitroresorcinol to the requirements for material (3.1) shall be determined by visual inspection.

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4.5 Test procedures.

4.5.1 Moisture. Transfer 3 grams of the sample to a tared weighing dish (6 cm in diameter). Break up any lumps with a spatula, cover the weighing dish and weigh the dish plus sample. Uncover the dish and heat at $68^{\circ} \pm 2^{\circ}C$ for 8 to 16 hours. Cool in a desiccator containing drierite or anhydrone and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100A}{W}$$

where A = loss in weight, grams
W = weight of sample, grams

Use this dried sample for the two assay determinations.

4.5.2 Assay by titration of nitro groups.

4.5.2.1 Apparatus. Storage and titration systems (see figs. 1 and 2).

4.5.2.2 Reagents.

4.5.2.2.1 Barium diphenylamine sulfonate solution. Dissolve 0.32 gram of barium diphenylamine sulfonate in 100 ml. of water.

4.5.2.2.2 Ammonium thiocyanate solution (20 percent). Dissolve 20 grams of NH_4CNS in 100 ml. of water.

4.5.2.2.3 Titanous chloride solution (0.2N). This solution can be prepared from either 20 percent titanous chloride solution or titanium hydride. To prepare it from 20 percent titanous chloride solution, proceed as follows: Mix 150 ml. of 20 percent titanous chloride solution with 100 ml. of hydrochloric acid and dilute to 1 liter with water. Mix with a current of inert gas (oxygen-free carbon dioxide or nitrogen). Filter through a Whatman No. 41 filter paper and if the solution is not clear, refilter through the same filter paper. Again mix with a stream of inert gas. Store in an amber bottle under a carbon dioxide atmosphere as shown in fig. 1.

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To prepare the titanous chloride solution from titanium hydride, proceed as follows: Heat 100 ml. of hydrochloric acid to 75° ± 5°C on a hot plate in a well ventilated hood. Add in small portions 12 grams of titanium hydride and cover with a watch glass. Remove the watch glass only to add titanium hydride to the solution. Warm until evolution of hydrogen has nearly ceased. Cool to room temperature, add 250 ml. of oxygen-free water, and mix for several minutes by means of a current of inert gas. Filter through a No. 41 Whatman filter paper and if the solution is not clear, refilter through the same paper. Add 100 ml. of hydrochloric acid and dilute to 1 liter with oxygen-free water. Again mix by a current of inert gas. Store in an amber bottle under carbon dioxide as shown in fig. 1.

4.5.2.2.3.1 Standardization. Standardize the titanous chloride solution as follows: Attach the source of inert gas to the titration flask as shown in fig. 1 and allow the gas to flow for 5 minutes to displace the air in the flask. Continue the flow of gas through the flask throughout the entire titration procedure. Transfer to the titration flask 0.2 to 0.3 gram of potassium dichromate (National Bureau of Standards No. 136) dried at 100°C for 2 hours, weighed to 0.1 mg. Add 50 ml. of 10 percent sulfuric acid solution. Titrate with 0.2 N titanous chloride solution until near the end point, as shown by the change in color to green (the change in color is from brownish purple, to purple, to green). Add 5 drops of barium diphenylamine sulfonate indicator solution and continue the titration until the end point is reached, as shown by the appearance of a light green color that does not change to a light blue color for at least 30 seconds (the light green color will change to light blue upon standing). Make a blank determination to correct for the presence of iron by adding 5 ml. of 20 percent ammonium thiocyanate solution to the above solution and then titrating with titanous chloride solution until the red color of the ferric thiocyanate completely disappears and the light blue color reappears. (If no iron is present, the solution will not turn red on the addition of the ammonium thiocyanate solution and the titration with titanous chloride will be unnecessary). Total the values obtained from the regular titration and the iron blank. Calculate the normality of the titanous chloride using the following equation:

$$\text{Normality of titanous chloride} = \frac{A}{0.04904B}$$

where A = weight of potassium dichromate, grams
B = titanous chloride solution (including iron blank), ml.

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4.5.2.2.4 Ferric ammonium sulfate solution (0.15 N standard solution). Pour 400 ml. of water into a 600 ml. beaker, add 75 grams of hydrated ferric ammonium sulfate ($Fe_2(SO_4)_3(NH_4)_2SO_4 \cdot 24H_2O$) and 25 ml. of sulfuric acid. Transfer the solution to a 1 liter graduate, dilute to capacity with water, and mix thoroughly by passing a current of air through the solution. Filter through a Whatman No. 1 filter paper. Standardize the solution as follows: Attach a source of inert gas to the titration flask as shown in fig. 1 and allow the gas to flow for 5 minutes to displace the air in the flask. Continue the flow of gas through the flask throughout the titration procedure. Transfer an accurately measured portion of 40 to 45 ml. of ferric ammonium sulfate solution to the titration flask (see fig. 1). Add 15 ml. of 15 percent hydrochloric acid solution and titrate with newly standardized 0.2 N titanous chloride solution until near the end point. Add 5 ml. of the 20 percent ammonium thiocyanate solution, and continue the titration until the red color disappears. Calculate the normality of the ferric ammonium sulfate using the following equation:

$$\text{Normality of ferric ammonium sulfate solution} = \frac{AN}{B}$$

where A = titanous chloride solution, ml.
N = normality of titanous chloride solution obtained in 4.5.2.2.3.1.
B = ferric ammonium sulfate solution, ml.

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4.5.2.3 Procedure. Transfer 0.8 gram of the dried sample (from the moisture determination), weighed to 0.1 mg., to a 100 ml. beaker. Add about 50 ml. of glacial acetic acid and stir with a stirring rod until dissolved. Transfer to a 250 ml. volumetric flask and dilute to the mark with glacial acetic acid[1]. Attach the source of inert gas (oxygen-free carbon dioxide or nitrogen) to the titration flask as shown in fig. 1 and allow the gas to flow for 5 minutes to displace the air. Continue the flow of gas during the entire titration procedure. Pipet an accurately measured 25 ml. aliquot of the solution of the sample into the titration flask (see fig. 2). Add 40.00 ml. of 0.2 N titanous chloride solution. Add 25 ml. of 15 percent hydrochloric acid and a few glass beads (to prevent bumping) and connect the flask to a reflux condenser. Place the flask and condenser on a hot plate and boil gently for 5 minutes. Without disconnecting the condenser, remove the flask from the hot plate. Increase the current of inert gas and cool the solution to room temperature in a cold water bath. Disconnect the condenser, add 5 ml. of 20 percent ammonium thiocyanate solution, and titrate with 0.15 N ferric ammonium sulfate solution to the end point, as indicated by the first permanent red color. Conduct a blank determination using exactly the same amount of reagents as above. This will provide a value for the amount of ferric ammonium sulfate solution equivalent to 40.00 ml. of titanous chloride solution. Calculate the percent trinitroresorcinol as follows:

$$\text{Percent trinitroresorcinol} = \frac{1.362\ (A-B)N}{W}$$

where A = ferric ammonium sulfate solution for blank, ml.
B = ferric ammonium sulfate solution for sample, ml.
N = normality of ferric ammonium sulfate solution
W = weight of sample in aliquot, grams.

[1]Glacial acetic acid has a high coefficient of expansion; therefore, to insure accuracy in the aliquot, the solution must be maintained at the same temperature until after the aliquot is taken.

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4.5.3 Assay by titration with standard alkali. Transfer 0.4 gram of the dried sample (from the moisture determination), accurately weighed to 0.1 mg., to a wide mouth 500 ml. Erlenmeyer flask. Add 150 ml. of warm water and swirl until dissolved. Cool to room temperature. Add 5 drops of 2 percent alcoholic phenolphthalein solution and titrate to a pink color with 0.1 N sodium hydroxide solution. Run a blank determination. Calculate the percent trinitroresorcinol as follows:

$$\text{Percent trinitroresorcinol} = \frac{12.26\ (A-B)N}{W}$$

where A = sodium hydroxide solution for titration of sample, ml.
B = sodium hydroxide solution for titration of blank, ml.
N = normality of sodium hydroxide solution.
W = weight of sample, grams.

4.5.4 Melting point. Dry a sample for the melting point determination, residue on ignition, water-insoluble matter, alcohol-insoluble matter, sulfate, and granulation by transferring about 30 gram portion to a large porcelain dish (or watch glass) and heating at 68° ± 2°C for 8 to 16 hours. Transfer to a stoppered bottle. Add a sample to a melting point capillary tube (0.9 to 1.2 mm in diameter and 10 cm high) (see 6.3.1) in 3 portions until the height of the trinitroresorcinol in the column is about 3-4 mm. Tap the capillary on the table top after each addition. Determine the melting point by the use of a Vanderkamp apparatus (or other suitable apparatus) (see 6.3.2). Regulate the electrical heating system so that the rise in temperature in the vicinity of the melting point is 1°C every 3 minutes. Observe the capillary tube through the magnifying glass and record as the melting point, the point at which the last solid particles have disappeared.

4.5.5 Residue on ignition. Transfer 3 grams of the dried sample to a tared platinum dish. Add 40 ml. of nitric acid and evaporate to dryness on the hot plate at moderate heat. Carry along a blank determination. Add 15 ml. of nitric acid and 3 ml. of sulfuric acid and evaporate to dryness on the hot plate at the highest heat. Heat over a Meker burner until the sulfuric acid has evaporated and then heat in a muffle furnace at 750°C for 15 minutes. Cool in a desiccator and weigh. Calculate the percent residue on ignition as follows:

$$\text{Percent residue on ignition} = \frac{100\ (A-B)}{W}$$

where A = weight of residue, grams
B = weight of residue in blank, grams
W = weight of sample, grams

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4.5.6 Water-insoluble matter. Transfer 2 grams of the dried sample to a 400 ml. beaker and add 300 ml. of water. Heat to boiling and stir until dissolved. Filter immediately through a tared sintered glass crucible of medium porosity and transfer and wash with hot water. Dry at 105°C for 1 hour, cool in a desiccator and weigh. Calculate the percent water-insoluble matter as follows:

$$\text{Percent water-insoluble matter} = \frac{100A}{W}$$

where A = weight of residue, grams
W = weight of sample, grams

4.5.7 Alcohol insoluble matter. Transfer 4 grams of the dried sample to a 250 ml. beaker. Add 120 ml. of 95 percent ethyl alcohol and stir with a stirring rod. Cover with a watch glass and heat to boiling on an electric hot plate with occasional stirring. Filter while hot through a tared sintered glass crucible of medium porosity. Transfer and wash with 95 percent ethyl alcohol Dry at 105°C for 30 minutes, cool in desiccator, and weigh. Calculate the percent alcohol-insoluble matter as follows:

$$\text{Percent alcohol-insoluble matter} = \frac{100A}{W}$$

where A = weight of residue, grams
W = weight of sample, grams

4.5.8 Sulfate. Transfer 5 grams of the dried sample to a 250 ml. beaker. Add 0.5 gram of sodium chloride, 25 ml. of hydrochloric acid, and 25 ml. of nitric acid. Carry along a blank determination. Evaporate to volume of about 1 to 2 ml. Add 10 ml. of hydrochloric acid and again evaporate to a volume of 1 to 2 ml. Add 5 ml. of hydrochloric acid and about 150 ml. of hot water. Filter through a Whatman No. 40 filter paper, wash with warm water, and discard the filter paper. Heat the filtrate to boiling, add 10 ml. of 10 percent barium chloride solution (10 percent), and boil for 1 or 2 minutes. Allow to stand overnight. Filter through a Whatman No. 42 filter paper and transfer and wash with 1 percent hydrochloric acid solution. Transfer the filter paper and precipitate to a tared platinum crucible and carefully char and burn off the filter paper at dull red heat. Finally ignite at red heat for 45 minutes. Cool in a desiccator and weigh. Calculate the percent sulfate (as H_2SO_4) as follows:

$$\text{Percent sulfate including resorcinol disulfonic acid (as } H_2SO_4) = \frac{42.02\ (A-B)}{B}$$

where A = weight of residue, grams
B = weight of residue in blank, grams
W = weight of sample, grams

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4.5.9 Granulation. Transfer 10 grams of the dried sample to a U. S. standard sieve No. 40 (conforming to the requirements of RR-S-366), 3 inches in diameter, to which a bottom pan has been attached. Shake for 3 minutes by hand or with a standard vibrating machine. If any residue remains on the sieve, brush it out into a tared weighing dish, dry at 68°C for 2 hours, cool in a desiccator, and weigh. Calculate the material that does not pass through the No. 40 sieve as follows:

$$\text{Material not passing through No. 40 sieve} = \frac{100A}{W}$$

where A = weight of residue, grams
W = weight of sample, grams

Note. Wash the sieve and pan well with water after finishing a test.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Trinitroresorcinol shall be packaged in a heat sealed polyethylene bag in accordance with PPP-C-301 inside a lined fiber drum conforming to PPP-D-723, type III, class 1. Each drum shall not exceed 200 pounds net (see 6.2).

5.1.2 Level B. Trinitroresorcinol shall be packaged in a heat-sealed polyethylene bag in accordance with PPP-C-301 inside a lined fiber drum conforming to PPP-D-723, type II, grade A. Each drum shall not exceed 100 pounds net (see 6.2).

5.1.3 Level C. Trinitroresorcinol shall be packed in shipping containers in accordance with PPP-C-301 for shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation (see 6.2).

5.2 Marking. In addition to any special marking specified, shipping containers shall be marked in accordance with MIL-STD-129.

5.2.1 Special labeling. Each container of trinitroresorcinol shall be labeled as follows:

TRINITRORESORCINOL
Do not take internally
Avoid breathing dust
Can cause skin irritation
Can be absorbed through the skin
Wash thoroughly after handling

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6. NOTES

6.1 Intended use. Trinitroresorcinol covered by this specification is intended for use in the manufacture of lead styphnate and as a constituent in some primers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required.
(c) Size and type of shipping container.
(d) Weight of trinitroresorcinol to be packed in a shipping container.
(e) Level of packing desired (see section 5).
(f) Special labeling (see 5.2.1).
(g) Moisture content of material to be shipped.

6.3 Sources of equipment.

6.3.1 Capillary tubes, Catalog No. 6401-2P, can be procured from Arthur H. Thomas, Philadelphia, Pa.

6.3.2 Vanderkamp apparatus, Catalog No. 63, M-1945, can be procured from Scientific Glass Apparatus Co., Inc., Bloomfield, N. J.

6.4 Sampling and testing precautions. Trinitroresorcinol when shocked or exposed to heat is an explosion hazard and should be handled with care.

Custodian:
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Preparing activity:
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Review activities:
Army - MU

Project No. 1376-A078

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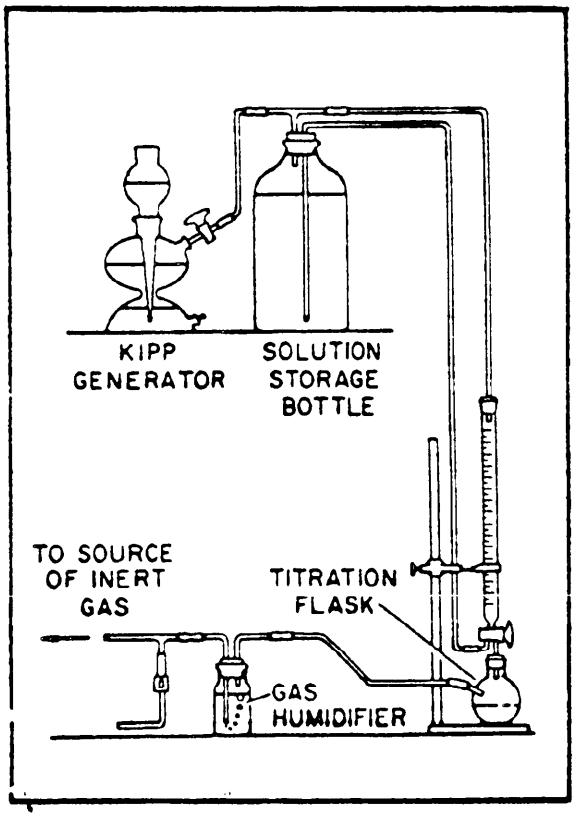


Figure 1. Storage System

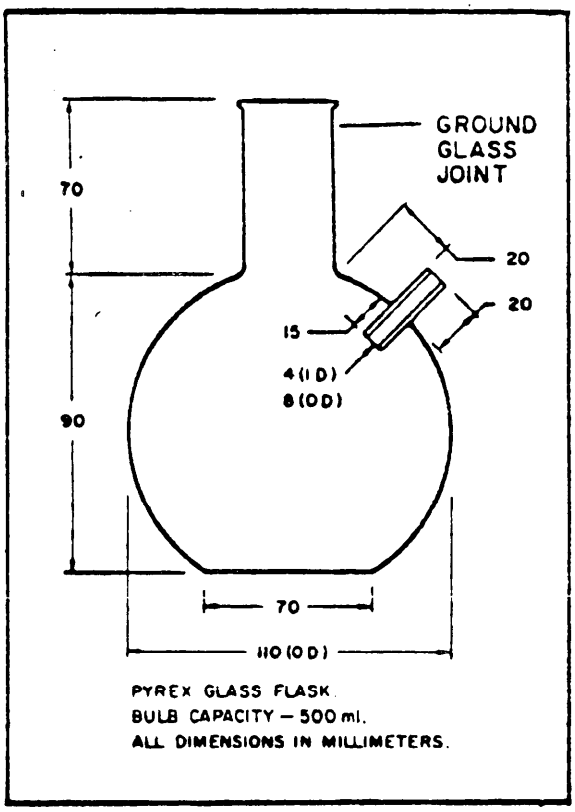


Figure 2. Titration Flask

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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